

Oleksandr Kolko

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PROFESSIONAL SUMMARY

Junior cloud and infrastructure engineer with **five months of commercial AWS experience**. During a security-startup internship I provisioned and maintained **up to five concurrent containerised lab environments** with Terraform (ECS and EKS on Fargate), built the CI/CD pipelines and kept every credential in Secrets Manager and KMS. Since then I have built keyless CI/CD platforms on GCP and Azure entirely as code and rebuilt an AWS pipeline as Terraform. The Azure build goes from an empty subscription to a live application in **under 12 minutes** with **no stored credential at any layer**. BSc Cybersecurity and Forensic Computing, **First Class Honours**, University of Portsmouth. My security background means least privilege, managed secrets and audit trails are where I start a build. Looking for a junior cloud, infrastructure or DevOps role.

CORE COMPETENCIES

AWS

Azure

GCP

Terraform

Kubernetes

Docker

CI/CD

Python

IAM & Least Privilege

WORK EXPERIENCE

CyberVectors Labs October 2025 - February 2026

Cybersecurity Engineering Intern

Remote

- Provisioned and maintained **up to 5 concurrent containerised lab environments** on AWS using Terraform (ECS and EKS on Fargate, backed by EC2 compute), managing configuration, dynamic port allocation, session isolation and uptime across all running instances.
- Built automated provisioning workflows integrating Python (FastAPI) and PHP (Laravel) backends with Docker APIs and AWS Lambda, reducing manual setup time and improving deployment consistency.
- Configured VPCs, security groups and network ACLs as Terraform-managed infrastructure to isolate lab environments from each other, and applied IAM least-privilege policies so users could only access their allocated resources.
- Managed secrets and credentials through **AWS Secrets Manager and KMS** rather than storing them in plaintext, and used S3 for lab environment data and artefact storage.
- Set up **CI/CD pipelines** using AWS CodePipeline, CodeBuild and CodeCommit with Git-based version control, giving the team consistent, auditable releases.
- Used CloudTrail to trace infrastructure issues back to their root cause and resolved them independently.
- Wrote technical documentation, deployment guides and architecture overviews used by both engineers and non-technical stakeholders.

Calvin Klein June 2024 - Present

Sales Assistant

Portsmouth

- Assisted 50+ customers daily in a fast-paced environment, adapting communication to resolve complex queries quickly.
- Managed inventory operations and worked within a team to meet operational targets consistently.

PROJECTS

Azure CI/CD Platform September 2026

Built a complete CI/CD platform on Azure **from an empty subscription, entirely as code**, the third build of the same pipeline after AWS console and AWS Terraform editions. Terraform across the azurearm, azuread and azuredevops providers provisions a VNet with delegated subnets and NSGs, **AKS with Azure CNI**, a private PostgreSQL Flexible Server with no public endpoint, ACR and Key Vault, with remote state in Azure Storage. **No stored credential at any layer, verified** (passwordCredentials: []): the pipeline authenticates through Entra ID workload identity federation, pods read secrets from Key Vault through the Secrets Store CSI driver, and the database password never enters Terraform state. Multi-stage Azure DevOps pipeline scripted in PowerShell, Ansible for manifests and database migrations, and Azure Monitor alerts verified by causing each condition. A point-in-time restore drill lost **zero data**. **Cold start to live application in 11m 44s**, push to live in 3m 20s, under GBP 1 for the whole project.

Terraform, AKS, Entra ID, Key Vault, Azure DevOps, PowerShell, Ansible, PostgreSQL, KQL

AWS CI/CD Pipeline, Terraform Edition

Rebuilt a hand-built AWS CI/CD pipeline as **33 Terraform resources across 13 files**, then destroyed the stack and rebuilt it from an empty account: a full apply takes about a minute and a push to main reaches production in roughly **100 seconds**. The rebuild surfaced **four defects** the console version had been hiding, including a deployment that reported seven successful lifecycle hooks while serving nothing. Wrote least-privilege IAM as policy documents up front, moved state to an encrypted, versioned S3 backend with native lockfile locking, used moved blocks for renames, and ran a drift experiment showing that a clean plan proves configured state, not intent.

Terraform, CodePipeline, CodeBuild, CodeDeploy, CodeArtifact, IAM, EC2

GCP/GKE DevSecOps Platform

Deployed a FastAPI service to a Terraform-provisioned, VPC-native GKE cluster on Spot nodes with Artifact Registry and GCS remote state. Keyless CI/CD in GitHub Actions gates on pytest, authenticates via Workload Identity Federation with **zero static service-account keys**, then builds, SHA-tags and helm upgrades on every merge. Detection-as-code alerts on service-account-key creation, with kube-prometheus-stack, Grafana and a GBP 10/month budget.

Terraform, GKE, Helm, GitHub Actions, Workload Identity Federation, Prometheus, Kafka

Anomaly Detection Proxy

Python reverse proxy that inspects inbound HTTP in real time, flagging rate-limit abuse, path probing and scanner user-agents, and logs and forwards rather than blocking to avoid disrupting legitimate users. Deployed on Kubernetes with 3 replicas, validated self-healing by deleting pods, and packaged with Helm using per-environment values files.

Python, Kubernetes, Helm, Docker, AWS S3

Machine Learning Phishing Detection Browser Extension Final-year project - 75%

Chrome/Firefox extension detecting phishing websites in real time with locally-run ML models, so no browsing data leaves the machine. Trained Random Forest and XGBoost models on approximately 8,000 URLs, achieving an **8% false positive rate**.

Python, XGBoost, Random Forest, Browser Extensions

EDUCATION

BSc Cybersecurity and Forensic Computing (BCS-accredited) University of Portsmouth September 2023 - July 2026

First Class Honours, dissertation graded **75%**. Malware Forensics **78%** (reverse engineering with IDA, static and dynamic analysis); Vulnerability Discovery **74%** (stack and heap overflows, format-string and integer bugs in gdb). Also: Ethical Hacking; Security Management (ISO 27001, NIST CSF, GDPR); Security and Cryptography; Forensics Fundamentals; Operating Systems and Internetworking; System Administration and Internetworking.

CERTIFICATIONS

AWS Certified Cloud Practitioner	Amazon Web Services	In progress
CompTIA Security+	CompTIA	In progress
CompTIA Network+	CompTIA	In progress
TryHackMe - Top 3%, 140+ rooms completed	TryHackMe	Ongoing

SKILLS

Cloud and Infrastructure: AWS hands-on across EC2, ECS, EKS, Fargate, Lambda, VPC, security groups, NACLs, S3, IAM, Secrets Manager, KMS and CloudTrail, provisioned as **Terraform-managed infrastructure during a real internship. Azure** from a full build: AKS, Entra ID, Key Vault, ACR, PostgreSQL Flexible Server, VNets, NSGs, private DNS, Log Analytics and Azure Monitor. GCP: GKE, Artifact Registry, Workload Identity Federation.

Infrastructure as Code and CI/CD: **Terraform** (remote state and locking, moved blocks, drift detection), Ansible, Helm. CI/CD in GitHub Actions, Azure DevOps and AWS CodePipeline, CodeBuild and CodeDeploy.

Containers: Extensive hands-on Docker. Kubernetes: deployments, replica scaling, self-healing, Services and Secrets, released via Helm with per-environment values files.

Scripting and Data: **Python** as primary language (FastAPI, automation, API integration). PowerShell, Bash. **SQL (PostgreSQL):** schema design, indexing, EXPLAIN ANALYZE, transactions and migrations. KQL.

Linux, Networking and Security: Ubuntu and Alpine day to day. TCP/IP, DNS, DHCP, VLANs, firewalls and VPCs. ISO 27001, NIST CSF and GDPR from the degree. Wireshark, Nmap, Burp Suite, WPScan, IDA, gdb, Autopsy, FTK Imager, Splunk, Git.